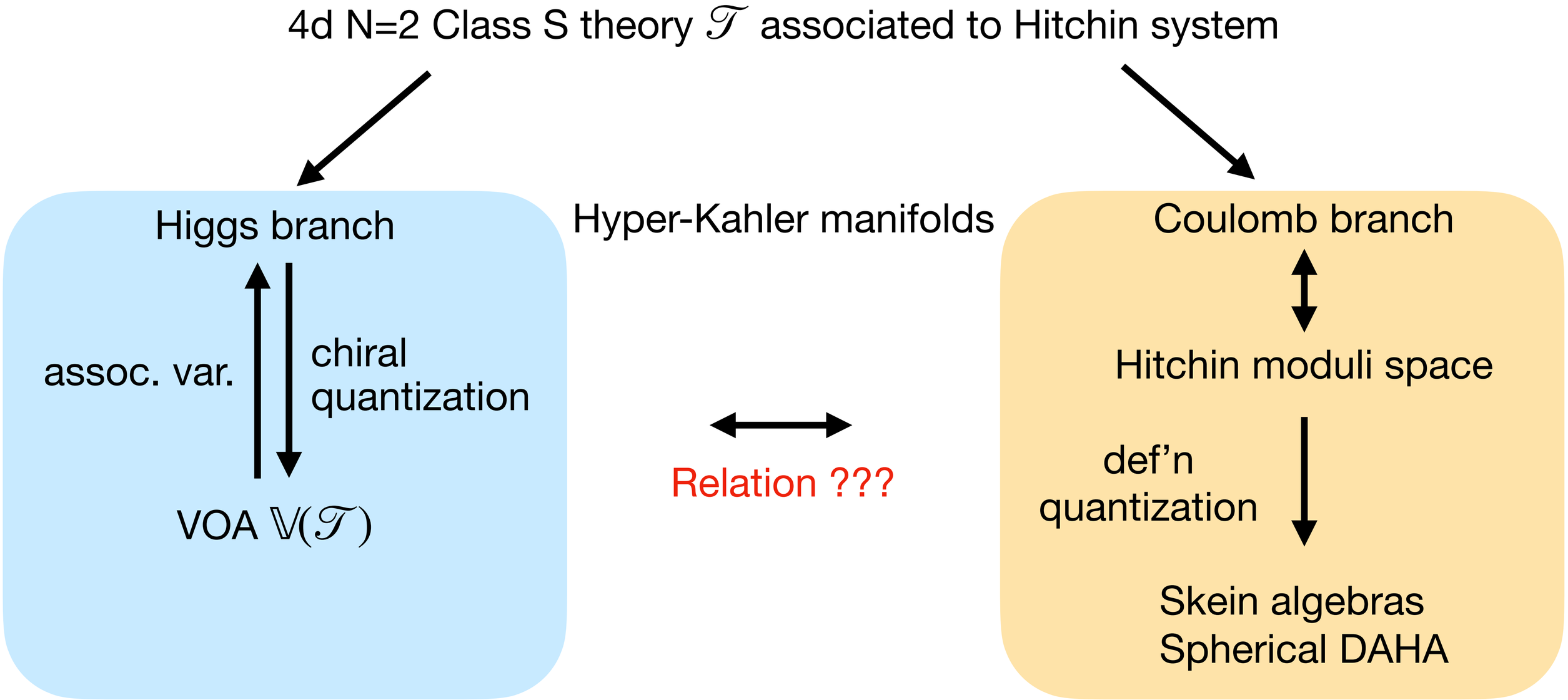
GRASP - November 2026

Mysterious relations between geometry of Hitchin moduli spaces and VOA modules from 4d N=2 theories

Satoshi Nawata Fudan University

Motivation


4d N=2 Class S theory $\mathscr{T}$ associated to Hitchin system
Higgs branch
assoc. var.
chiral quantization
VOA $\mathbb{V}(\mathscr{T})$
Hyper-Kahler manifolds
Relation ???
Coulomb branch
Hitchin moduli space
def'n quantization
Skein algebras
Spherical DAHA

Class S theory

6d (2,0) SCFT of type $\mathfrak{g}$ ADE

$$\downarrow \quad \mathbb{R}^3 \times S^1 \times C_{g,n}$$

Gaiotto

Gaiotto-Moore-Neitzke

4d N=2 Class S theory $\mathcal{T}_{g,n}$ associated to Hitchin system

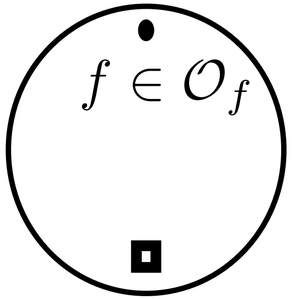
- Regular class S theory ($g \geq 0,\ n \geq 0$ s.t. $2g-2+n>0$)



$$\phi|_{z_i} \sim \frac{f_i^\vee}{z-z_i} + \cdots, \quad f_i \in \mathcal{O}_{f_i}$$

Higgs branch = Star-shaped quiver variety

- Argyres-Douglas type (a sphere with one irregular and one regular puncture)



$$\phi|_0 \sim \frac{f^\vee}{z} + \cdots,$$

$$\phi|_\infty \sim \frac{s}{w^{2+\frac{u}{m}}} + \cdots, \qquad s\text{: semisimple regular}$$

Dan Xie

Disclaimer

My contribution to this field is negligible, but I am very much interested in this subject

Argyres-Douglas, Boundary Admissible Level, Isolated fixed points

- L. Fredrickson, D. Pei, W. Yan, K. Ye, arXiv:1701.08782
- L. Fredrickson, A. Neitzke, arXiv:1709.06142
- C. Kozçaz, S. Shakirov, W. Yan, arXiv:1801.08316
- M. Dedushenko, S. Gukov, H. Nakajima, D. Pei, K. Ye, arXiv:1809.04638
- S. Gukov, P. Koroteev, S. Nawata, D. Pei, I. Saberi, arXiv:2206.03565
- P. Shan, D. Xie, W. Yan, arXiv:2306.15214, arXiv:2404.00760

Class S, Non-Admissible Level, fixed manifolds

- Y. Pan, W. Yan, arXiv:2410.15695, arXiv:2412.03155
- S. Gukov, P. Hsin, S. Park, D. Pei, arXiv:2511.13696
- D. Xie, arXiv:2509.20872, arXiv:2606.16714
- P. Shan, W. Yan, Q. Zhao, arXiv:2507.09254, arXiv:2606.08990, arXiv:2608.00428
- Y. Li, Y. Pan, W. Yan, arXiv:2510.03888

Refer to Wenbin Yan's talk at ICBS 2025

Disclaimer

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The (very rough) upshot is

Simple modules of $\mathbb{V}(\mathscr{T}(C_{g,n}, G))$ $\overset{\text{1-1}}{\longleftrightarrow}$ $\mathbb{C}^\times$-fixed components of $\mathscr{M}_H(C_{g,n}, G^\vee)$

As far as I understand, some statements are proven, but this correspondence is still at the observational level

This talk is part of my own learning process

Comments, suggestions and questions are welcome during the talk.

SECTION 2

Preliminary: SCFT/VOA correspondence

SCFT/VOA Correspondence

Beem-Lemos-Liendo-Peelaers-Rastelli-van Rees,

4d $\mathcal{N}=2$ SCFT $\mathcal{T} \longrightarrow$ VOA $\mathbb{V}(\mathcal{T})$

- The 2d central charge is related to the 4d central charge by

$$c_{2d} = -12 c_{4d}$$

- The vacuum character of the VOA equals the Schur index of the 4d theory:

$$\chi_{\mathbb{V}(\mathcal{T})} = I_{\text{Schur}}\left(\mathcal{T}\right)$$

- The flavor symmetry algebra decomposes into simple factors

$$\mathfrak{g}_F = \bigoplus_i \mathfrak{g}_i \oplus \mathfrak{u}(1)^{\oplus s},$$

and each simple factor gives an affine Lie subalgebra $L_{k^{2d}}(\mathfrak{g}_i) \subset \mathbb{V}(\mathcal{T})$, with negative level

$$k_i^{2d} = -k_i^{4d}$$

- The associated variety $X_{\mathbb{V}(\mathcal{T})}$ is the Higgs branch.

Affine vertex algebras and their associated varieties

- Zhu's C_2-algebra: $R_{\mathbb{V}} = \mathbb{V}/C_2(\mathbb{V})$, $C_2(\mathbb{V}) = \mathrm{span}\{a_{(-2)}b \mid a, b \in \mathbb{V}\}$
  with modes $Y(a,z) = \sum_{n\in\mathbb{Z}} a_{(n)} z^{-n-1}$
  commutative associative Poisson algebra: $a \cdot b = a_{(-1)}b$, $\{a, b\} = a_{(0)}b$

- Associated variety: $X_{\mathbb{V}} := (\mathrm{Spec}\, R_{\mathbb{V}})_{\mathrm{red}}$ (Poisson variety)

- $\mathbb{V}$ is quasi-lisse if $X_{\mathbb{V}}$ has finitely many symplectic leaves
  $\Rightarrow$ finitely many simple ordinary modules, modular invariant characters

- For affine VOAs: $X_{V^k} = \mathfrak{g}^* \cong \mathfrak{g}$, $X_{L_k} \subset \mathfrak{g}$ closed, G-stable, conic
  L_k quasi-lisse $\iff$ $X_{L_k} \subseteq \mathcal{N}$

  - $k \notin \mathbb{Q}$ or $k < -h^\vee$: $X_{L_k} = \mathfrak{g}$
  - $k = -h^\vee$: $X_{L_k} = \mathcal{N}$
  - admissible k ($k + h^\vee = p/q$, $p \geq h^\vee$; $\mathfrak{g}$ simply-laced): $X_{L_k} = \overline{\mathbb{O}}$ (nilpotent orbit closure)
  - non-admissible rational k: interesting but mysterious, known only in small families

Kac

Frenkel

Zhu

Arakawa

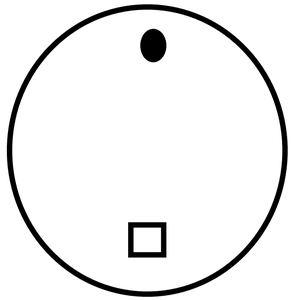
...

SECTION 3

VOA modules vs $\mathbb{C}^\times$-fixed points of $\mathscr{M}_H$:

Boundary admissible level

(A_1, D_3) Argyres-Douglas theory



$$\phi(w) \sim \frac{\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)}{w} + \cdots$$

$$\mathbb{V}(A_1, D_3) = L_{-4/3}(\mathfrak{sl}_2)$$

3 simple modules: $\{0,\ -\frac{2}{3}\,\omega,\ -\frac{4}{3}\,\omega\} \circlearrowleft SL(2,\mathbb{Z})$

$$\phi(z) \sim \frac{\left(\begin{smallmatrix} 1 & \\ & -1 \end{smallmatrix}\right)}{z^{2+\frac{1}{2}}} + \cdots$$

Bezrukavnikov-Boixeda Alvarez-McBreen-Yun

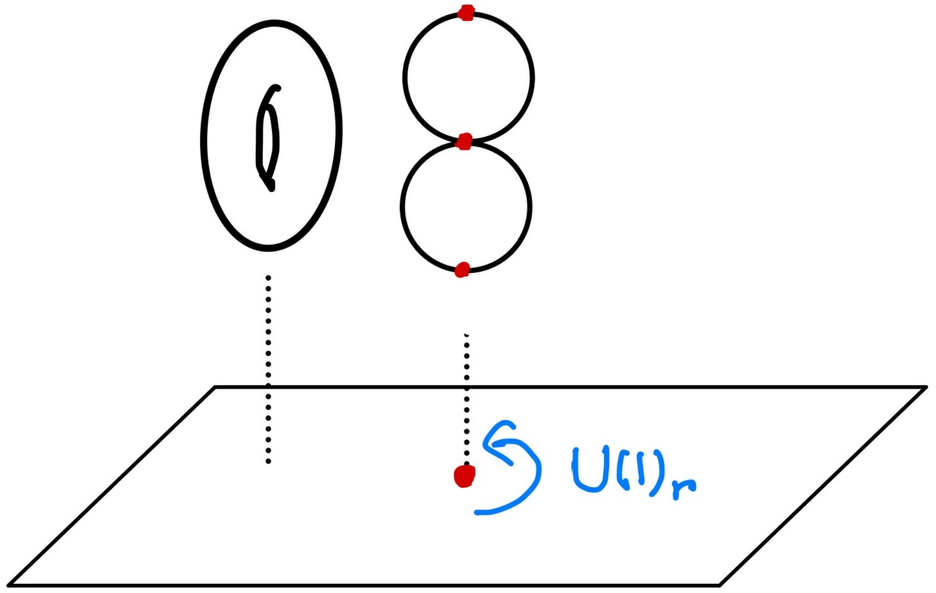
3 fixed points in the central fiber (Kodaira type III)

In wild Hitchin moduli space

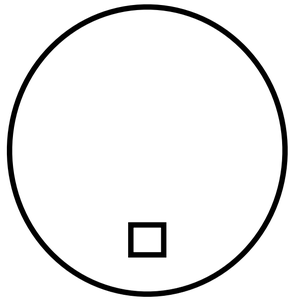
the central fiber is affine Springer fiber $\gamma = \left(\begin{smallmatrix} 0 & 1 \\ z^3 & 0 \end{smallmatrix}\right)$

$\mathbb{C}^\times \circlearrowleft Fl_\gamma = \{g \in G((z))/I \mid Ad_{g^{-1}}(\gamma) \in \text{Lie } I\}$

Kodaira III



(A_1, A_2) Argyres-Douglas theory



$$\phi(z) \sim \frac{\begin{pmatrix} 1 & \\ & -1 \end{pmatrix}}{z^{2+\frac{3}{2}}} + \cdots$$

$\mathbb{V}(A_1, A_2) =$ Lee-Yang minimal model

2 primaries $\{\varphi_0, \varphi_{-\frac{2}{5}}\} \circlearrowleft PSL(2, \mathbb{Z})$

Bezrukavnikov-Boixeda Alvarez-McBreen-Yun

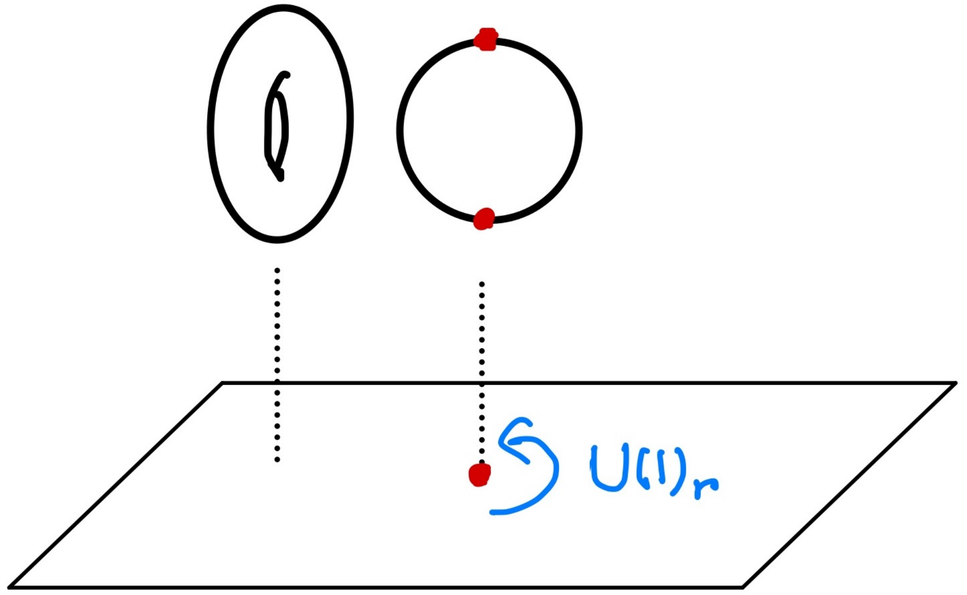
2 fixed points in the central fiber (Kodaira type II)

In wild Hitchin moduli space

the central fiber is affine Spaltenstein fiber $\gamma = \begin{pmatrix} 0 & 1 \\ z^3 & 0 \end{pmatrix}$

$\mathbb{C}^\times \circlearrowleft Gr_\gamma = \{g \in G((z))/G[[z]] \mid Ad_{g^{-1}}(\gamma) \in \mathfrak{g}[[z]]\}$

Kodaira II



Double Affine Hecke Algebra

- Rank-one DAHA:

$$\ddot{H}(\mathbb{Z}_2) = \mathbb{C}_{q,t}[T^{\pm 1}, X^{\pm 1}, Y^{\pm 1}] \Big/ \left\{ \begin{array}{ll} TXT = X^{-1}, & Y^{-1}X^{-1}YXT^2 = q^{-1}, \\ TY^{-1}T = Y, & (T-t)(T+t^{-1}) = 0 \quad \text{(Hecke)} \end{array} \right\}$$

- $SL(2,\mathbb{Z})$ action:

$$\begin{array}{lccl} \tau_+ : & (X,Y,T) & \mapsto & (X,\ q^{-\frac{1}{2}}XY,\ T) \\ \tau_- : & (X,Y,T) & \mapsto & (q^{\frac{1}{2}}YX,\ Y,\ T) \\ \sigma : & (X,Y,T) & \mapsto & (Y^{-1},\ XT^2,\ T) \end{array}$$

- Non-symmetric Macdonald polynomials E_j (generic q,t) from intertwiners

$$\mathsf{A} := XT, \qquad \mathsf{B} := t\Big(T + \frac{t-t^{-1}}{Y^{-2}-1}\Big)$$

$$1 = E_0 \overset{\mathsf{A}}{\longleftrightarrow} E_1 \overset{\mathsf{B}}{\longleftrightarrow} E_{-1} \overset{\mathsf{A}}{\longleftrightarrow} E_2 \overset{\mathsf{B}}{\longleftrightarrow} E_{-2} \overset{\mathsf{A}}{\longleftrightarrow} E_3 \overset{\mathsf{B}}{\longleftrightarrow} E_{-3} \overset{\mathsf{A}}{\longleftrightarrow} \cdots$$

- At $t^2q^3 = 1$, there exists a 3-dim irrep of $\ddot{H}(\mathbb{Z}_2)$, which receives a $SL(2,\mathbb{Z})$ action

Spherical Double Affine Hecke Algebra

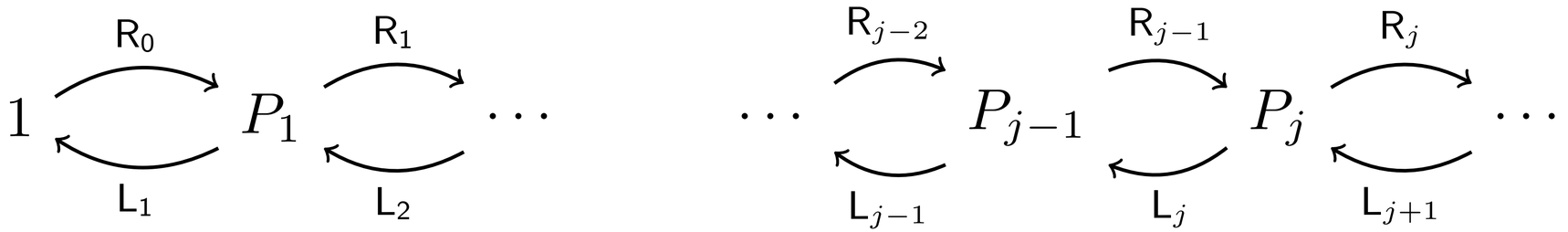
- Spherical DAHA: $S\ddot{H}(\mathbb{Z}_2) := e\,\ddot{H}(\mathbb{Z}_2)\,e, \quad e = \dfrac{T+t^{-1}}{t+t^{-1}} \quad (e^2 = e,\ Te = t\,e)$
  generated by $x = (X + X^{-1})\,e,\ \ y = (Y + Y^{-1})\,e,\ \ z$ with relations

$$[x,y]_q = (q^{-1}-q)\,z, \qquad [y,z]_q = (q^{-1}-q)\,x, \qquad [z,x]_q = (q^{-1}-q)\,y,$$
$$q^{-1}x^2 + qy^2 + q^{-1}z^2 - q^{-\frac{1}{2}}xyz = (q^{-\frac{1}{2}}t - q^{\frac{1}{2}}t^{-1})^2 + (q^{\frac{1}{2}} + q^{-\frac{1}{2}})^2$$

- $PSL(2,\mathbb{Z})$ action:

$$\begin{aligned}
\tau_+ :&\quad (x,y,z) \quad\mapsto\quad \left(x,\ \frac{xy+yx}{q^{1/2}+q^{-1/2}} - z,\ y\right)\\
\tau_- :&\quad (x,y,z) \quad\mapsto\quad \left(\frac{xy+yx}{q^{1/2}+q^{-1/2}} - z,\ y,\ x\right)\\
\sigma :&\quad (x,y,z) \quad\mapsto\quad \left(y,\ x,\ \frac{xy+yx}{q^{1/2}+q^{-1/2}} - z\right)
\end{aligned}$$

- Polynomial representation on $\mathbb{C}_{q,t}[X+X^{-1}]$: symmetric Macdonald polynomials P_j

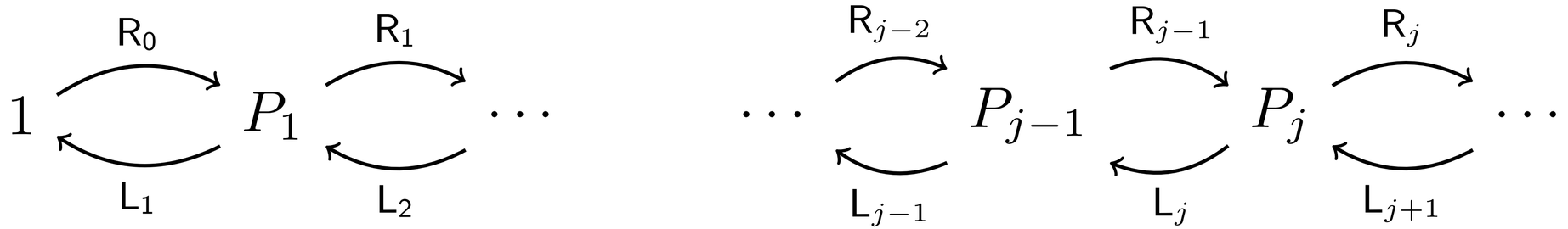


Spherical Double Affine Hecke Algebra

- $PSL(2,\mathbb{Z})$ action:

$$\begin{aligned}
\tau_+ : \quad (x,y,z) &\mapsto \left(x,\ \frac{xy+yx}{q^{1/2}+q^{-1/2}} - z,\ y\right)\\
\tau_- : \quad (x,y,z) &\mapsto \left(\frac{xy+yx}{q^{1/2}+q^{-1/2}} - z,\ y,\ x\right)\\
\sigma : \quad (x,y,z) &\mapsto \left(y,\ x,\ \frac{xy+yx}{q^{1/2}+q^{-1/2}} - z\right)
\end{aligned}$$

- Polynomial representation on $\mathbb{C}_{q,t}[X+X^{-1}]$: symmetric Macdonald polynomials P_j



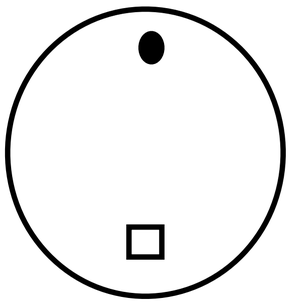
- At $t^2q^3=1$: e projects the 3-dim irrep M of $\ddot{H}(\mathbb{Z}_2)$ to a 2-dim rep eM of $S\ddot{H}(\mathbb{Z}_2)$, which receives a $PSL(2,\mathbb{Z})$ action

SL(2,Z) action

$$
\begin{array}{ccc}
K^0\big(\text{ Rep } L_{-\frac{4}{3}}(\mathfrak{sl}_2)\big) & \cong & \ddot{H}(\mathbb{Z}_2)\ \text{f.d.-mod}_{t^2q^3=1} \\
\circlearrowleft & & \circlearrowleft \\
SL(2,\mathbb{Z}) & & SL(2,\mathbb{Z})
\end{array}
$$

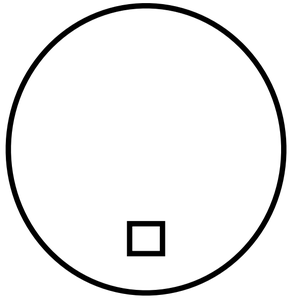
$$
\begin{array}{ccc}
K^0\big(\text{ Rep } \mathrm{Vir}(2,5)\big) & \cong & S\ddot{H}(\mathbb{Z}_2)\ \text{f.d.-mod}_{t^2q^3=1} \\
\circlearrowleft & & \circlearrowleft \\
PSL(2,\mathbb{Z}) & & PSL(2,\mathbb{Z})
\end{array}
$$

Closure of regular puncture



$$\phi(w) \sim \frac{\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)}{w} + \cdots$$

$$\phi(z) \sim \frac{\left(\begin{smallmatrix} 1 & \\ & -1 \end{smallmatrix}\right)}{z^{2+n+\frac{1}{2}}} + \cdots$$



$$\phi(z) \sim \frac{\left(\begin{smallmatrix} 1 & \\ & -1 \end{smallmatrix}\right)}{z^{2+n+\frac{1}{2}}} + \cdots$$

(A_1, D_{2n+3}) theory (A_1, A_{2n}) theory

$L_{-2+\frac{2}{2n+3}}(\mathfrak{sl}_2)$ qDS Vir$(2,2n+3)$

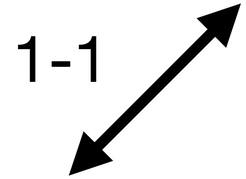
$(2n+3)$-dim $\ddot{H}$-irrep Idempotent projection $2n$-dim $S\ddot{H}$-irrep

Web of relations

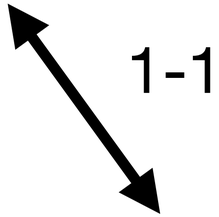
Simple modules of affine Lie algebras $L_k(\mathfrak{g})$

with boundary admissible level

Peng Shan-Dan Xie-Wenbin Yan







$\mathbb{C}^\times$-fixed points of

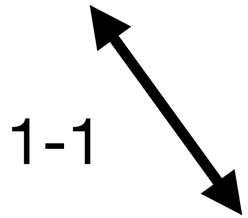
affine Springer fiber $Fl_\gamma(\mathfrak{g}^\vee)$

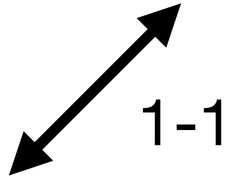
$PSL(2,\mathbb{Z})$

Finite-dimensional module

of DAHA $t^{h^\vee}q^n = 1$

$q^m = 1$





Bezrukavnikov-Boixeda Alvarez-McBreen-Yun

$\mathbb{C}^\times$-fixed points of wild Hitchin moduli space with $G^\vee$

But,

But, Why?

SECTION 4

VOA modules vs $\mathbb{C}^\times$-fixed manifolds of $\mathscr{M}_H$:

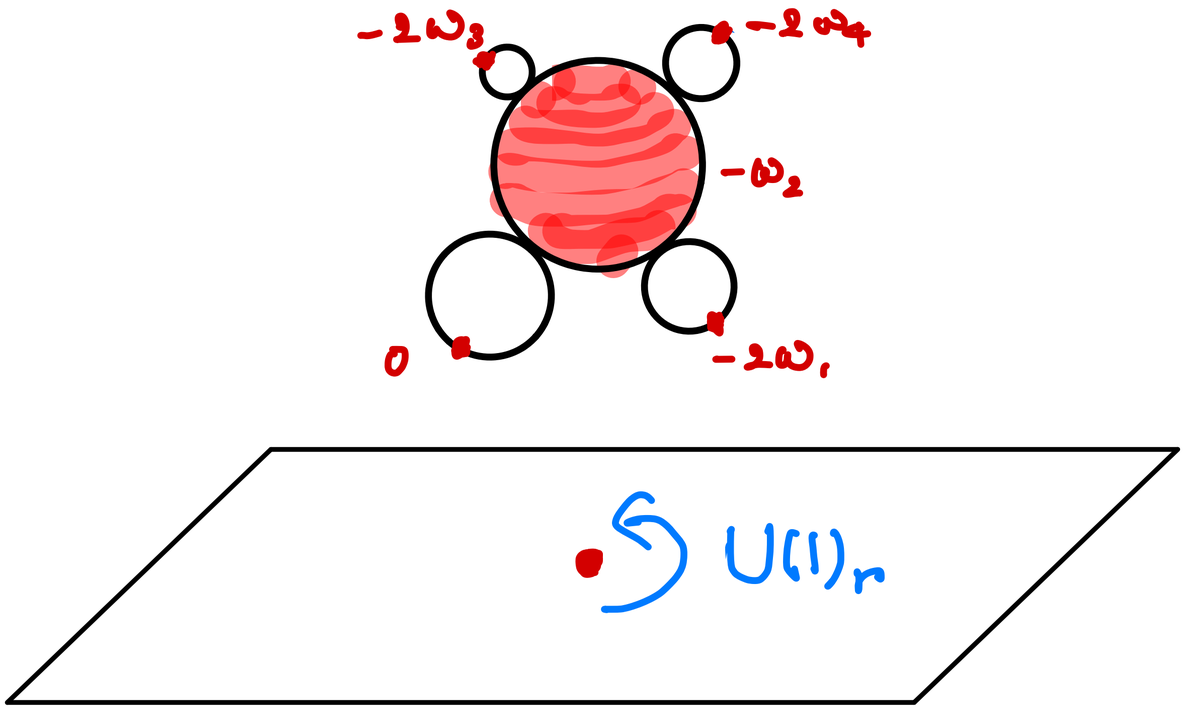
Non-admissible level

4-punctured sphere

- For a 4-punctured sphere, VOA is affine D_4 Lie algebra $\mathbb{V}(\mathcal{T}_{0,4}) = L_{-2}(\mathfrak{d}_4)$ with level -2
- There are 5 simple modules:

$$0\ ,\ -2\omega_1\ ,\ -2\omega_3\ ,\ -2\omega_4\ ,\ -\omega_2$$

- Fixed components under the $U(1)_r$ action consists of 4 points and $\mathrm{Bun}_G \cong \mathbb{P}^1$:



Fixed components vs simple modules

- The $\mathbb{V}(\mathcal{T}_{0,4})$-module characters $\{\chi_\lambda\}$ satisfy a set of (flavor) modular differential equations (FMDEs) These form a set of partial differential equations. e.g.

$$\mathrm{ch}''(\tau) - \frac{1}{6} E_2\, \mathrm{ch}'(\tau) - \frac{35}{144} E_4\, \mathrm{ch}(\tau) = 0$$

- The quasi-modularity of FMDEs implies that characters belong to a representation $\mathbb{S}_{0,4}^{mod}$ of the modular group

- Generally, the dimension of $\mathbb{S}_{0,4}^{mod}$ is larger than the number of SMs, where the extra solutions are usually logarithmic modules, and T is not semisimple.

$$SL(2,\mathbb{Z}) \circlearrowright \mathbb{S}_{0,4}^{mod} = \mathrm{Span}_{\mathbb{C}}\{\, 5 \text{ characters} + \log\text{-module} \,\}\ , \qquad \mathrm{Jord}(T) = [2,\, 1^4]$$

- Another perspective comes from the mixed Hodge polynomial of character variety We need its pure part:

$$\mathrm{PH}\big(\mathcal{M}_{0,4}^{\mathrm{Betti}}\,;\, q\big) = 4 + q$$

- # of simple modules = # fixed manifolds = # Jordan block $(T) = PH(q = 1)$

Fixed components vs simple modules

Yiwen Pan-Wenbin Yan

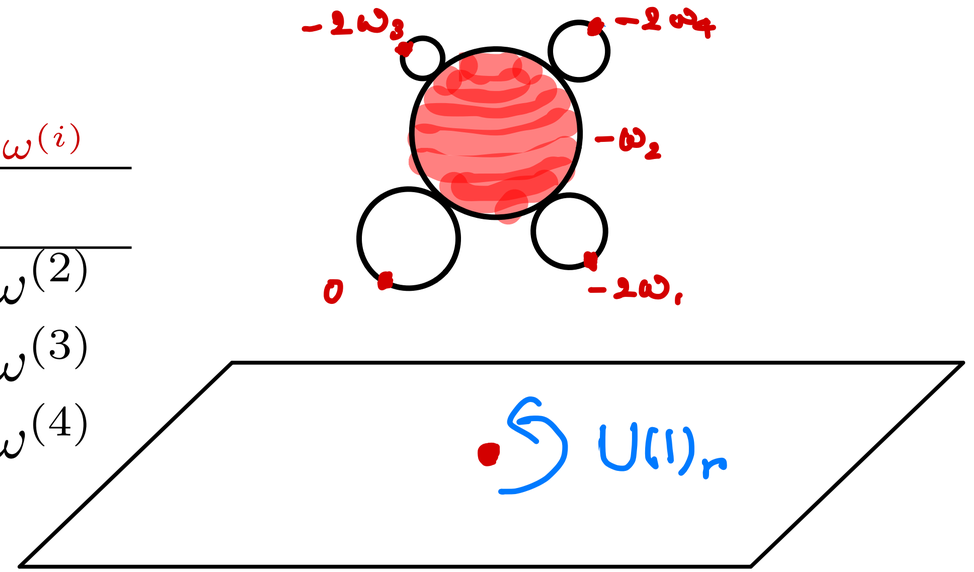
- $$SL(2,\mathbb{Z}) \circlearrowright \mathbb{S}^{mod}_{0,4} = \mathrm{Span}_{\mathbb{C}}\left\{ 5\text{characters} + \text{log-module} \right\}, \qquad \mathrm{Jord}(T) = [2, 1^4]$$

- Another perspective comes from the mixed Hodge polynomial of character variety We need its pure part:

$$\mathrm{PH}\left(\mathcal{M}^{\mathrm{Betti}}_{0,4}; q\right) = 4 + q$$

- # of simple modules = # fixed manifolds = # Jordan block $(T) = PH(q=1)$

$L(\omega_i)$	h	$\mathfrak{sl}(2)^{\otimes 4}$ rep	$\dim M$	$\mu - \mu_{max}\vert_{\alpha_i \mapsto -\omega^{(i)}}$
$-\omega_2$	-1	$[-2,0,0,0]$	1	$-1-2\omega^{(1)}$
$-2\omega_1$	-1	$[-2,-2,0,0]$	0	$-1-2\omega^{(1)}-2\omega^{(2)}$
$-2\omega_3$	-1	$[-2,0,-2,0]$	0	$-1-2\omega^{(1)}-2\omega^{(3)}$
$-2\omega_4$	-1	$[-2,0,0,-2]$	0	$-1-2\omega^{(1)}-2\omega^{(4)}$
$-2\omega_0$	0	$[0,0,0,0]$	0	0



1-punctured torus



- Moduli space of flat $\mathrm{SL}(2,\mathbb{C})$-connections on a once-punctured torus C_p

$$\pi_1(C_p) = \langle \mathfrak{m}, \mathfrak{l}, \mathfrak{c} | \mathfrak{m}\mathfrak{l}\mathfrak{m}^{-1}\mathfrak{l}^{-1} = \mathfrak{c} \rangle \ .$$

- These variables are subject to the cubic equation

$$\mathcal{M}_{\mathrm{flat}}(C_p, \mathrm{SL}(2,\mathbb{C})) = \{(x,y,z) \in \mathbb{C}^3 | x^2 + y^2 + z^2 - xyz - 2 = t^2 + t^{-2}\} \ .$$

  From the 4-punctured sphere, we can obtain it by setting $\theta_{1,2,3} = 0$.

- The deformation quantization of $\mathscr{O}(\mathfrak{X})$ with respect to Ω_J is the spherical DAHA $S\ddot{H}_{q,t}$ of type A_1

$$\mathscr{O}(\mathfrak{X}) \rightsquigarrow \mathscr{O}^q(\mathfrak{X}) \cong S\ddot{H}_{q,t}$$

- Relations are given by:

$$\begin{aligned}
[x,y]_q &= (q^{-1} - q)z \\
[y,z]_q &= (q^{-1} - q)x \\
[z,x]_q &= (q^{-1} - q)y \\
q^{-1}x^2 + qy^2 + q^{-1}z^2 - q^{-\frac{1}{2}}xyz &= (q^{-\frac{1}{2}}t - q^{\frac{1}{2}}t^{-1})^2 + (q^{\frac{1}{2}} + q^{-\frac{1}{2}})^2 \ ,
\end{aligned}$$

Symmetry

- Modular group $\mathrm{PSL}(2,\mathbb{Z})$: Since the algebra is constructed from a torus, it enjoys $\mathrm{PSL}(2,\mathbb{Z})$-automorphism generated by

$$\tau_+ \mapsto \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \qquad \tau_- \mapsto \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

and

$$\sigma := \tau_+\tau_-^{-1}\tau_+ = \tau_-^{-1}\tau_+\tau_-^{-1} \mapsto \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \tau_+\tau_-^{-1} \mapsto \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix} .$$

The actions on the generators are given by

$$\begin{aligned} \tau_+ : &\quad (x,y,z) \mapsto \left(x, \frac{xy+yx}{q^{1/2}+q^{-1/2}} - z, y \right), \\ \tau_- : &\quad (x,y,z) \mapsto \left(\frac{xy+yx}{q^{1/2}+q^{-1/2}} - z, y, x \right), \\ \sigma : &\quad (x,y,z) \mapsto \left(y, x, \frac{xy+yx}{q^{1/2}+q^{-1/2}} - z \right). \end{aligned}$$



- "sign changes" $\Xi = \mathbb{Z}_2 \times \mathbb{Z}_2$ are holomorphic involutions, and its generators ξ_1, ξ_2 and their combination $\xi_3 := \xi_1 \circ \xi_2$ act as

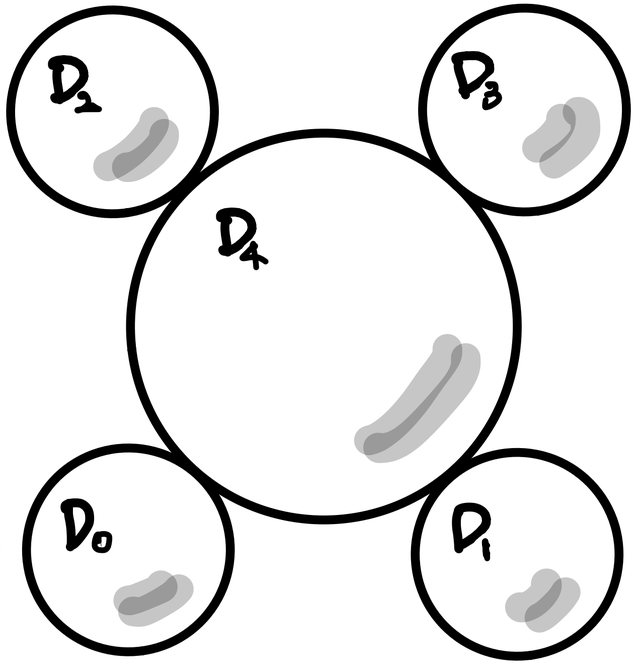
$$\begin{aligned} \xi_1 : &\quad (x,y,z) \mapsto (-x,y,-z) , \\ \xi_2 : &\quad (x,y,z) \mapsto (x,-y,-z) , \\ \xi_3 : &\quad (x,y,z) \mapsto (-x,-y,z) , \end{aligned}$$

Brane quantization

PSL(2,Z)-actions on $\mathfrak{B}_{\mathbf{D}_0}$ and $\mathfrak{B}_{\mathbf{D}_4}$ arises from modular tensor categories

Virasoro minimal model

$PSL(2,\mathbb{Z}) \circlearrowright \mathrm{Hom}(\mathfrak{B}_{D_0}, \mathfrak{B}_{\mathrm{cc}})$



Affine Lie algebra $L_k(\mathfrak{a}_1)$

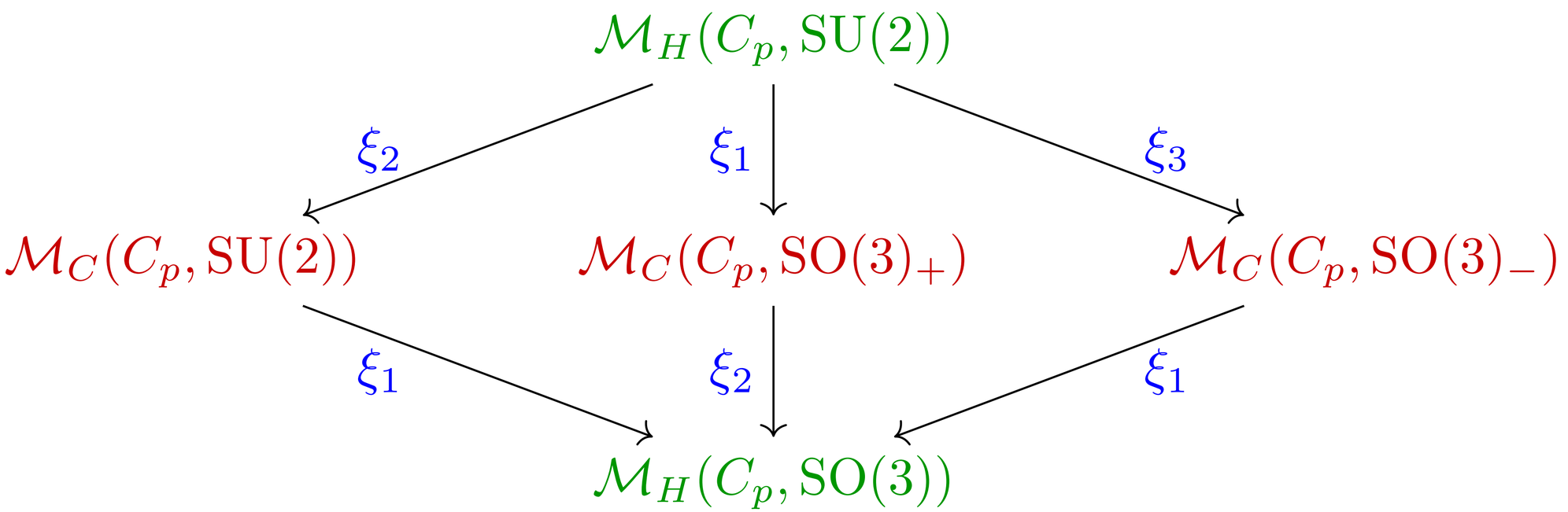
$PSL(2,\mathbb{Z}) \circlearrowright \mathrm{Hom}(\mathfrak{B}_{D_4}, \mathfrak{B}_{\mathrm{cc}})$

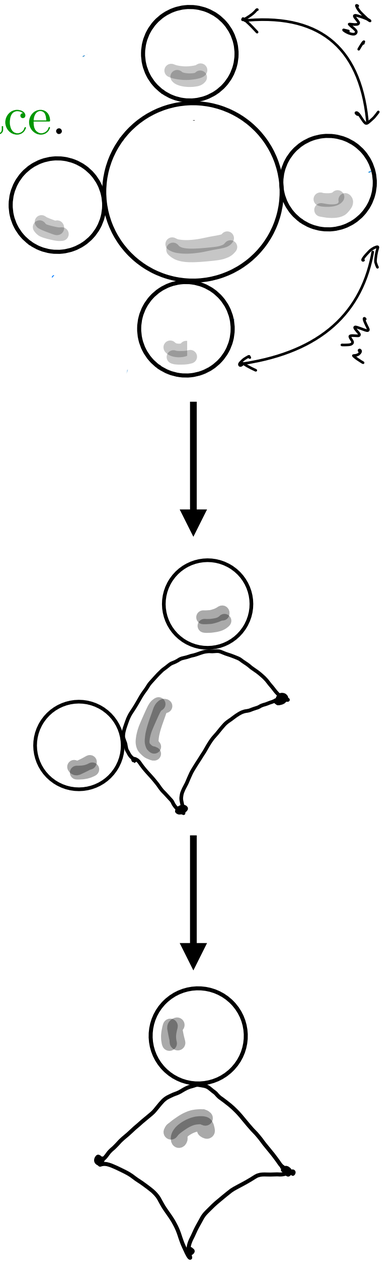
$(2,2\ell+1)$ Virasoro minimal model is VOA of $(A_1, A_{2(\ell-1)})$ Argyres-Douglas theory

Refer to my talk for this topic.

Hitchin moduli space and Coulomb branch

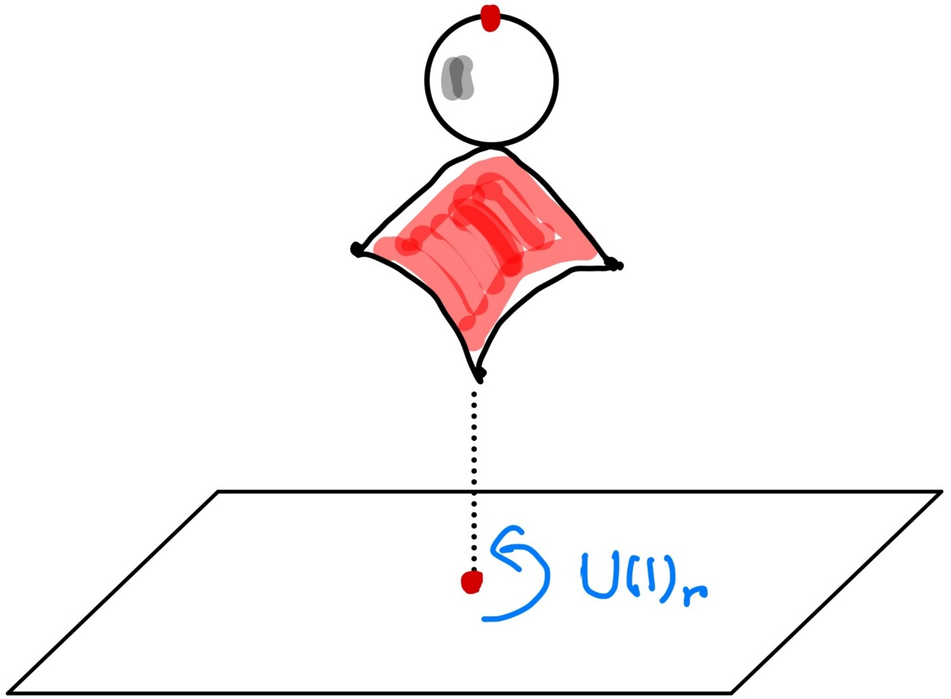
Coulomb branch of 4d $\mathcal{N}=4$ theory is a $\mathbb{Z}_2$ quotient of SU(2) Hitchin moduli space.





Simple modules vs fixed manifolds via Langlands duality

- For a once-punctured torus, VOA $\mathbb{V}(\mathcal{T}_{0,1})$ is small $\mathcal{N}=4$ superconformal Virasoro algebra with $c=-9$.
- There are 2 simple modules of $\mathbb{V}(\mathcal{T}_{0,1})$: $0, -\omega_1$.
- $\mathbb{S}_{1,1}^{\mathrm{mod}} = \mathrm{Span}_{\mathbb{C}}(2 \text{ simple} + \log)$, Jordan type of T is [2,1]
- Fixed manifolds of SO(3) Hitchin moduli space $\mathcal{M}_H(C_p, \mathrm{SO}(3))$ consist of a point $+ \mathbb{P}^1/(\mathbb{Z}_2 \times \mathbb{Z}_2)$
- SU(2) $\leftrightarrow$ SO(3) Langlands duality Gukov-Pei-Hsin



We have a dictionary

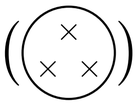
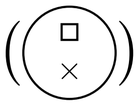
Simple modules of $\mathbb{V}(\mathscr{T}(C_{g,n}, G))$ $\overset{\text{1-1}}{\longleftrightarrow}$ $\mathbb{C}^\times$-fixed components of $\mathscr{M}_H(C_{g,n}, G^\vee)$

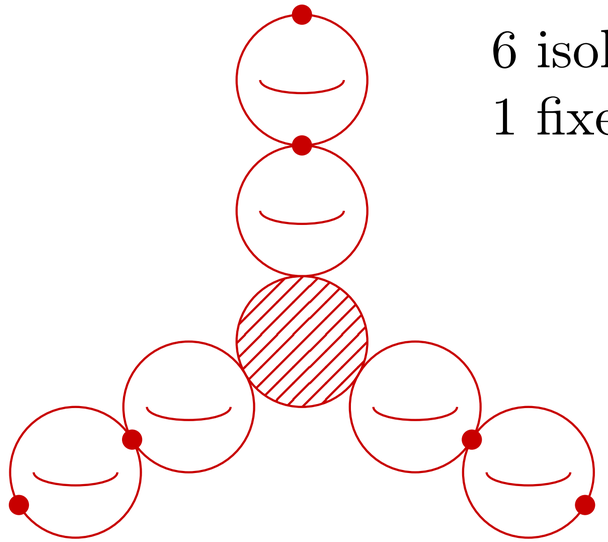
We have a dictionary

Simple modules of $\mathbb{V}(\mathscr{T}(C_{g,n}, G))$ $\overset{\text{1-1}}{\longleftrightarrow}$ $\mathbb{C}^\times$-fixed components of $\mathscr{M}_H(C_{g,n}, G^\vee)$

But, Why?

E6 Minahan-Nemeschansky theory

- $\mathcal{T} = T_3$ (E_6 Minahan–Nemeschansky)
- VOA: $\mathbb{V}(\mathcal{T}) = L_{-3}(E_6)$
- 7 simple modules [Arakawa–Moreau]
  + 1 log-module [Arakawa–Kawasetsu]
- $\mathcal{M}_C(\mathcal{T}) \simeq \mathcal{M}^{A_2}_{\mathrm{Hit}}$  $\simeq \mathcal{M}^{E_6}_{\mathrm{Hit}}$ 
- $\mathbb{C}^\times$-fixed points: 6 points + 1 $\mathbb{P}^1$



6 isolated pts
1 fixed $\mathbb{P}^1$ (shaded)

Kodaira type IV* (affine E_6)

Puzzle from 4d N=3 $\mathbb{Z}_3$ S-fold theory

Z. Guo-SN-Y. Pan-Q. Zhang

	T_3 (E_6 MN, $\mathcal{N}=2$)	rank-1 $\mathbb{Z}_3$ S-fold ($\mathcal{N}=3$)
singular fibre of $\mathcal{M}_C$	IV^*	IV^*
flavor symmetry	E_6	$U(1)$
VOA	$L_{-3}(E_6)$	$\mathcal{W}_{\mathbb{Z}_3}$
simple + log modules	$7+1=8$	$4+1=5$
$\mathbb{C}^\times$-fixed locus	6 pts + 1 $\mathbb{P}^1$	?
$\dim H^*$(fixed locus)	$6+2=8$	?

- Puzzle: the fibre IV^* alone predicts 8, but the VOA gives 5
- The Kodaira type does not see the flavor symmetry or the $\mathcal{N}=3$ structure
- Open: what is the correct $\mathbb{C}^\times$-geometry for the S-fold?
  How does the $U(1)$ flavor symmetry enter on the Coulomb-branch side?

Mysterious relationship

Simple modules of $\mathbb{V}(\mathscr{T}(C_{g,n}, G))$ $\overset{\text{1-1}}{\longleftrightarrow}$ $\mathbb{C}^\times$-fixed components of $\mathscr{M}_H(C_{g,n}, G^\vee)$

- More precise correspondence has been formulated for G=SU(2) Yiwen Pan-Wenbin Yan
- Very little is understood for higher-rank
- Koszul type duality? What is the underlying intrinsic reason?
- Very mysterious even for physicists
- Any insights from mathematicians will be appreciated